

Landing-Only Linear Search for an Escaping Target: Discovery and Capture

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Abstract. We study online search on the real line by a unit-speed flying searcher for a target that starts at unknown distance $D \geq 1$ from the origin and moves directly away from the origin at constant speed $v < 1$. Unlike classical linear search, the searcher cannot observe the target while flying: detection is possible only at a landing, and only when the target lies between the origin and the landing point. We distinguish *discovery*, where the process stops when the target is detected, from *capture*, where the searcher must subsequently meet the target. When v is known we give tight bounds for both objectives. When the initial distance is known but v is not, a doubly-exponential landing strategy achieves competitive ratio $O((1 - v)^{-23/4})$ for both objectives as $v \rightarrow 1^-$, and we conjecture that this exponent is optimal.

Keywords: Linear search · escaping target · competitive ratio · online algorithms · mobile agents

JC: This is a skeleton assembled from the two drafts in `references/`. Sections marked `TODO` still need to be written or verified; see `notes/model_and_results.md` for the current status of each claim.

1 Introduction

Online search asks how efficiently a mobile agent can locate an object whose relevant parameters are not known in advance. The classical one-dimensional problem, often called linear search or the cow-path problem [3,2], places a stationary target at an unknown point of the line and compares an online zig-zag trajectory with a clairvoyant searcher. The optimal deterministic competitive ratio is 9, achieved by doubling the excursion length on alternating sides.

Table 1: Summary of results. “ D known” means the magnitude $|d| = D$ is known but the sign of d is not. Entries marked $\dagger$ have an upper bound whose matching lower bound is not yet written in full.

Objective	Information	Competitive ratio	Status
Discovery	$v = 0, D$ unknown	$\frac{11+5\sqrt{5}}{2} \approx 11.09$	upper bound; lower bound †
Capture	$v = 0, D$ unknown	$\frac{25}{2}$	upper bound; lower bound †
Discovery	v known, D unknown	Eq. (8)	upper bound; lower bound †
Capture	v known, D unknown	Eq. (11)	upper bound; lower bound †
Discovery	D known, v unknown	$O((1-v)^{-23/4})$	upper bound
Capture	D known, v unknown	$O((1-v)^{-23/4})$	upper bound

The present paper studies a mobile analogue with two additional constraints. First, the target is *escaping*: it moves away from the origin at a fixed speed $v < 1$. Second, the searcher is a flying agent with *landing-only* visibility. It receives no information while in flight. A target can be discovered only at a landing, and only if it lies on the segment joining the origin to the landing point. In this sense the searcher is “forward blind”: overshooting the target in flight does not reveal it.

We distinguish two objectives. In the *discovery* problem the cost is the time of the first landing that reveals the target. In the *capture* problem the searcher must, after discovery, physically meet the target.

1.1 Related work

Demaine, Fekete, and Gal studied linear and star search with a fixed turn cost [11]; Bose and De Carufel developed a general framework for searching on a line [4,5]. Search for an oblivious target moving away from the origin was introduced in [10], and the effect of knowing the target’s speed or initial distance was studied in [8,9]. The information regimes considered there motivate the four-way classification used here. Our model differs in a crucial sensing assumption: passing the target in flight does not reveal it. The landing-only restriction is also the one-dimensional analogue of the planar search-and-rescue model of [7], where an agent must land on the x -axis before it can search along it.

JC: TODO: expand related work; cite the search-games books [1,12] and the search-and-rescue line [6].

1.2 Contributions and organization

Section 2 gives the model. Section 3 treats the stationary target ($v = 0$). Section 4 gives tight bounds when v is known and D is unknown. Section 5 treats the case where D is known but v is not. Section 6 records a conjectured matching lower bound exponent for the latter case. Table 1 summarizes the results.

2 Model and notation

A searcher starts at the origin of $\mathbb{R}$ and moves with maximum speed 1. A target starts at

$$d = \sigma D, \quad \sigma \in \{-1, +1\}, \quad D \geq 1,$$

and moves directly away from the origin at a constant speed $0 \leq v < 1$, so its position at time t is $y(t) = \sigma(D + vt)$.

The searcher cannot detect the target while flying. At a landing at position x at time t , it discovers the target if x has the correct sign and $|x| \geq D + vt$; equivalently, the target lies between the origin and the landing point.

A *landing strategy* is specified by alternating landing points $x_i = (-1)^i a_i$ with $a_i > 0$, $i \geq 0$. Write

$$S_i := \sum_{j=0}^i a_j, \quad S_{-1} := 0.$$

Because the searcher returns through the origin between consecutive landings, the i th landing occurs at time

$$t_i = 2S_{i-1} + a_i = 2S_i - a_i. \quad (1)$$

The *effective reach* of landing i against speed v is

$$B_i(v) := a_i - vt_i = (1 - v)a_i - 2vS_{i-1}, \quad (2)$$

and a target on the side of landing i is discovered there exactly when $D \leq B_i(v)$.

If σ , D , and v are all known, a unit-speed searcher meets the target at time

$$T^*(D, v) = \frac{D}{1 - v}, \quad (3)$$

and a clairvoyant searcher can land exactly at the meeting point, so we use (3) as the benchmark for both discovery and capture. For known v and unknown D the competitive ratio of a strategy $\mathcal{A}$ is $\sup_{\sigma, D} T_{\mathcal{A}}(\sigma D, v) / T^*(D, v)$. When D is known but v is not, we use the speed-dependent ratio

$$\text{CR}_{\mathcal{A}}(v) := \sup_{\sigma \in \{-1, +1\}} \frac{T_{\mathcal{A}}(\sigma D, v)}{D/(1 - v)},$$

and by scaling we may normalize $D = 1$.

Suppose the target is discovered at landing i . The searcher is then beyond the target and reverses direction, closing at relative speed $1 + v$, so the total capture time is

$$T_i^{\text{cap}}(D, v) = t_i + \frac{a_i - D - vt_i}{1 + v} = \frac{2S_i - D}{1 + v}. \quad (4)$$

3 Stationary target

JC: TODO. Both drafts cover $v = 0$: geometric excursions $a_i = r^i$ give discovery ratio $r^2(r+1)/(r-1)$, minimized at $r = (1+\sqrt{5})/2$ with value $(11+5\sqrt{5})/2$, and capture ratio $2r^3/(r-1) - 1$, minimized at $r = 3/2$ with value $25/2$. EK's notes ([references/dis-cap-line.pdf](#), Sections 2.2.2 and 2.2.3) start the matching lower bounds via the $s_i = S_{i-1}/x_i$ recurrence but stop at the recurrence inequalities; those arguments need to be completed.

4 Known speed, unknown distance

Assume v is known, whereas $D \geq 1$ and σ are unknown. Consider geometric excursions $a_i = r^i$ with $r > 1$, so $S_i = (r^{i+1} - 1)/(r - 1)$. The effective reach (2) is eventually positive only if

$$r > \frac{1+v}{1-v}, \quad (5)$$

since $B_i(v)/a_i \rightarrow ((1-v)r - (1+v))/(r-1)$.

4.1 Discovery

A target on the side of landing i that is first discoverable there was missed at landing $i-2$, so the worst case is $D \downarrow B_{i-2}(v)$ with discovery cost t_i . Asymptotically in i ,

$$\text{CR}_{\text{disc}}(r, v) = (1-v) \frac{r^2(r+1)}{(1-v)r - (1+v)}. \quad (6)$$

Theorem 4.1 (Known-speed discovery). *For landing-only discovery with unknown initial distance and known speed v , the optimal geometric expansion is*

$$r_{\text{disc}}^*(v) = \frac{1 + 2v + \sqrt{5 + 4v}}{2(1-v)}, \quad (7)$$

and the resulting competitive ratio is

$$\text{CR}_{\text{disc}}^*(v) = \frac{(\sqrt{5+4v} + 3)(2v + \sqrt{5+4v} + 1)^2}{4(1-v)^2(\sqrt{5+4v} - 1)}. \quad (8)$$

At $v = 0$ this is $(11 + 5\sqrt{5})/2$, and $\text{CR}_{\text{disc}}^*(v) \sim 27/(1-v)^2$ as $v \rightarrow 1^-$.

Proof. Minimize $f(r) = r^2(r+1)/((1-v)r - (1+v))$ over the feasible range (5). Then $f'(r) = 0$ if and only if $(1-v)r^2 - (1+2v)r - (1+v) = 0$, whose feasible root is (7); substituting into (6) gives (8). These identities are checked symbolically in `code/verify_known_speed.wls`.

JC: TODO: the lower bound. The draft in [references/landing_only_mobile_line_search.pdf](#) reduces it to the “critical recurrence” $S_i + S_{i-1} = K((1-v)S_{i-2} - (1+v)S_{i-3})$ and asserts, without proof, that the sharp constant equals the geometric minimum. We need an actual argument (e.g. the s_i monotone-sequence technique from EK's notes, or the Gal/Bose–De Carufel framework).

4.2 Capture

With the same worst case $D \downarrow B_{i-2}(v)$, equation (4) gives

$$\text{CR}_{\text{cap}}(r, v) = \frac{1-v}{1+v} \left(\frac{2r^3}{(1-v)r - (1+v)} - 1 \right). \quad (9)$$

Theorem 4.2 (Known-speed capture). *The optimal geometric expansion for capture is*

$$r_{\text{cap}}^*(v) = \frac{3(1+v)}{2(1-v)}, \quad (10)$$

and the resulting competitive ratio is

$$\text{CR}_{\text{cap}}^*(v) = \frac{27(1+v)}{2(1-v)^2} - \frac{1-v}{1+v} = \frac{(2v+1)(v+5)^2}{2(1-v)^2(1+v)}. \quad (11)$$

At $v = 0$ this is $25/2$, and $\text{CR}_{\text{cap}}^*(v) \sim 27/(1-v)^2$ as $v \rightarrow 1^-$.

Proof. It suffices to minimize $K(r) = r^3/((1-v)r - (1+v))$. The derivative vanishes when $3r^2((1-v)r - (1+v)) = (1-v)r^3$, i.e. $2(1-v)r = 3(1+v)$, giving (10) and $K^* = 27(1+v)^2/(4(1-v)^3)$. Substituting into (9) yields (11). See `code/verify_known_speed.wls`.

JC: TODO: lower bound, as for discovery.

EK: The formulas for $\text{CR}_{\text{disc}}(v)$ and $\text{CR}_{\text{cap}}(v)$ in [references/dis-cap-line.pdf](#) (Table 1 and Sections 2.3.2–2.3.3) differ from (8) and (11) for $v > 0$ although they agree at $v = 0$. The simulator `code/simulate.py` agrees with the formulas above; see [notes/model_and_results.md](#).

5 Known distance, unknown speed

Assume $D = |d|$ is known (normalize $D = 1$) but σ and v are unknown. No uniform constant ratio is possible as $v \rightarrow 1^-$, so we bound the ratio as a function of the actual speed.

5.1 Doubly-exponential strategy and induced speed thresholds

Fix $A > 1$ and $b > 1$ and use alternating excursions

$$a_i = A^{b^i}. \quad (12)$$

The maximal speed handled by round i is defined by the critical equality $1 + v_i t_i = a_i$, so by (1)

$$v_i = \frac{a_i - 1}{a_i + 2S_{i-1}}, \quad \delta_i := 1 - v_i = \frac{1 + 2S_{i-1}}{a_i + 2S_{i-1}}. \quad (13)$$

Since the sequence is doubly exponential, $S_i = a_i(1 + o(1))$ and $a_{i-1}/a_i = A^{-(b-1)b^{i-1}}$, hence

$$\delta_i \sim 2 \frac{a_{i-1}}{a_i} = 2A^{-(b-1)b^{i-1}}. \quad (14)$$

For a fixed actual speed v let $i(v) := \max\{i : v_i < v\}$. Round $i(v) + 1$ is speed-sufficient; since the sign is unknown, the adversary can force one more round, so discovery happens by round $i(v) + 2$.

Theorem 5.1 (Unknown-speed discovery). *For the strategy (12),*

$$\text{CR}_{\text{disc}}(v) = O((1-v)^{-\alpha(b)}), \quad \alpha(b) := \frac{b^3}{b-1} - 1,$$

as $v \rightarrow 1^-$. The exponent is minimized uniquely at $b = 3/2$, giving $\text{CR}_{\text{disc}}(v) = O((1-v)^{-23/4})$.

Proof. JC: TODO: port the proof from the draft (Theorem 5.1 there). Sketch: with $i = i(v)$, $\delta_{i+1} \leq 1 - v < \delta_i$; the normalized cost δt_{i+2} is increasing in δ , so the worst case on this interval is $\delta_i a_{i+2} \asymp A^{[b^3 - (b-1)]b^{i-1}}$ while $\delta_i^{-1} \asymp A^{(b-1)b^{i-1}}$. Then $\alpha'(b) = b^2(2b-3)/(b-1)^2$.

Theorem 5.2 (Unknown-speed capture). *For the strategy (12), $\text{CR}_{\text{cap}}(v) = O((1-v)^{-\alpha(b)})$ with the same $\alpha(b)$, hence $O((1-v)^{-23/4})$ at $b = 3/2$.*

Proof. JC: TODO. By (4) with $D = 1$, $\text{CR}_{\text{cap}}(v) \leq \frac{1-v}{1+v}(2S_{i(v)+2} - 1) \leq (1 + o(1))(1-v)a_{i(v)+2}$, and the exponent calculation is identical to Theorem 5.1.

6 A conjectured matching lower exponent

Normalize $D = 1$ and consider an arbitrary alternating landing strategy with $\delta_i = (1 + 2S_{i-1})/(S_i + S_{i-1})$ as in (13). When $S_i \gg S_{i-1}$, $\delta_i \asymp S_{i-1}/S_i$. Taking a speed just above v_i and the adverse sign, both discovery and capture cost roughly $\delta_i S_{i+2}$ in normalized terms. With local growth exponents $p_i := \log S_i / \log S_{i-1}$, this suggests the local lower-bound exponent $p_i p_{i+1} p_{i+2} / (p_i - 1) - 1$, which for constant p is $p^3 / (p - 1) - 1$, minimized at $p = 3/2$ with value $23/4$.

Conjecture 6.1 (Universal unknown-speed lower exponent). For every deterministic landing-only strategy $\mathcal{A}$ there is a sequence $v_k \rightarrow 1^-$ such that, for discovery and likewise for capture,

$$\text{CR}_{\mathcal{A}}(v_k) = \Omega((1 - v_k)^{-23/4 + o(1)}).$$

7 Conclusion and open problems

JC: TODO. Open directions from the drafts: (1) prove or refute Conjecture 6.1; (2) complete the lower bounds for the known-speed and stationary cases; (3) nonstationary doubly-exponential schedules (varying b) to improve lower-order factors; (4) randomized strategies; (5) positive landing times or turn costs, connecting to [11].

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